(a) segment efficiency (every solver at its efficiency-first working point)

(b) segment false rate

clean segment efficiency

segment false rate

moderate segment efficiency

segment false rate

heavy segment efficiency

segment false rate

tracks per event T

tracks per event T

classical

1BQF

QSVT $d = 40$, base A

QSVT $d = 40$, A + occupancy $\alpha = 0.05$

QSVT $d = 40$, A + bifurcation $\beta = 0.5$

QSVT $d = 40$, A + occupancy + bifurcation